Data Structures and Algorithms

Imperial College Business School, Business Analytics

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Overview

This module introduces computational problem solving through the design of algorithms and data structures. Devising efficient methods for working with data is at the core of most modern technological innovation, from search engines to healthcare and finance. We build algorithms in Python, moving from an ambiguous problem statement to a working computational solution: covering searching, sorting, core data structures, object-oriented programming, and graph algorithms, before finishing with a look at problems that resist efficient solutions altogether (the knapsack problem and greedy algorithms).

No previous programming experience is assumed, but the module moves quickly, so some early self-study helps if you're new to programming (see below).

Structure

The module runs as **ten two-hour classes**. Each class combines a short lecture with hands-on Python exercises, supported by slides and the activity notebook. Most classes include a short graded activity, due about a week later; see the separate **Assessment** document for details.

> **This module is hybrid in format.** Each class's activity notebook opens with a short **Preparation** section (readings and reflection questions). Go through it beforehand, so you come to class having already done some of the thinking.

Prerequisites and recommended materials

If you'd like a head start, spend a few hours on the first two chapters of an online tutorial such as *Automate the Boring Stuff with Python* before the module begins.

The recommended textbook is:

- *Introduction to Computation and Programming Using Python* (3rd edition, 2021), John Guttag, MIT Press. We cover roughly the first half of the book.

Session-specific readings, videos, and further references are linked directly within each class's own materials, rather than listed here.

Class schedule

1. Introduction to Python
2. Algorithms and functions
3. Sequential data
4. Computational complexity and search algorithms
5. Sorting algorithms
6. Data structures, files, and errors
7. OOP and data structures
8. Graph algorithms on unweighted graphs
9. Graph algorithms on weighted graphs
10. Greedy algorithms and the knapsack problem

Getting started

See `setup/getting-started.pdf` (included with your course materials on Canvas) for setting up your Python environment before Class 1.

Getting help

If you get stuck or have questions, ask in class: time is set aside at the end of each class for open questions.

Assessment

Grading, due dates, and collaboration policy are covered in the separate **Assessment** document.